
Few-Step Flow-Matching Error Can Be Misranked by Averaged Jacobian Regularity: A Certified Gaussian Counterexample

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Abstract

1 Does averaged squared Jacobian regularity rank two interpolants in the same
2 order as equal-NFE endpoint error? Chen et al. propose minimizing A_2 , the time-
3 integrated squared spatial Jacobian norm of a flow-matching marginal ODE, as
4 a path-selection criterion, without a proved discretization-error bound. Already
5 for independent $N(0, 1)$ and $N(0, 4)$, the exact integrals are $5\pi/8 - 1$ (linear) and
6 $\pi^2/16$ (trigonometric VP), while Heun at NFE 8 reverses Gaussian W_2 . The linear
7 factor is the rational $6797469/3559400$; a rational enclosure in $\mathbb{Q}[\pi, \sqrt{2}]$ gives
8 $r_{VP} < 187/100$. The object is a flow-matching marginal ODE, not a score-based
9 probability-flow ODE. The Gaussian drift and W_2 formula are closed form; the
10 regularity integrand is exact, while multimode $\mathcal{R}$ is evaluated deterministically
11 by adaptive quadrature. Pairwise and four-path census comparisons are reported
12 below; a finite census does not imply that a global A_2 -minimizer minimizes fixed-
13 NFE error.

14 1 Introduction

15 Few-step sampling commits to a path, a solver, and a small NFE budget before any error is observed.
16 Lipschitz constants enter classical one-step bounds [Hairer et al., 1993]. Chen et al. [2025] propose
17 minimizing A_2 , the time-integrated squared spatial Jacobian norm (their Definition 3.2), as a *selection*
18 criterion. They prove no bound relating A_2 to a discretization error. We test whether the pairwise
19 order that A_2 induces between two interpolant paths is reliable at equal NFE on a flow-matching
20 marginal ODE.

21 We isolate that ranking question on commuting Gaussian interpolants, where the field is affine. The
22 Gaussian drift and W_2 formula are closed form; the regularity integrand is exact, while multimode
23 $\mathcal{R}$ is evaluated deterministically by adaptive quadrature. The leading result is a deterministic scalar
24 Heun example (Section 3). A modal identity then records the signed information that $\mathcal{R}$ discards
25 (Section 4). Section 5 splits the ranking question into three comparisons: pairwise linear versus VP,
26 pairwise VP versus the Chen Ex. 3.3 scalar schedule, and a four-path census including the per-mode
27 path, which is not a C^1 shared interpolant when Σ_1 has at least two distinct eigenvalues. Section 6
28 records a complementary Class S construction in which the $\mathcal{R}$ -minimizer is Euler-inexact.

29 Contemporaneous work studies Lipschitz Euler rates [Stéphanovitch, 2026], sharpness-aware
30 grids [Gupta et al., 2026], least-action paths [Du et al., 2026], and instance-aware discretizations [Yuan
31 et al., 2026]. Those results control other objects. Tao and Choi [2026] give the general strain/vorticity
32 form of the phenomenon; Section 7 records the precise relation.

2 Gaussian interpolants and averaged regularity

Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent. A shared schedule is a scalar pair (α, σ) . We distinguish three regularity classes.

- Class S: $\alpha, \sigma \in C^1([0, 1])$ with $\alpha(0) = \sigma(1) = 1$ and $\alpha(1) = \sigma(0) = 0$. Linear and trigonometric VP belong to Class S.
- Class Q: a C^1 path of SPD covariances $Q(t)$ with continuous $A = \frac{1}{2}Q'Q^{-1}$. The per-mode schedule $q_i(t) = \lambda_i^t$ is Class Q.
- Class R: $\alpha^2, \sigma^2 \in C^1([0, 1])$, $\alpha^2 \geq 0$, $\sigma^2 \geq 0$, and $\alpha(0)^2 = 1$, $\sigma(0)^2 = 0$, $\alpha(1)^2 = 0$, $\sigma(1)^2 = 1$. Square roots need not be C^1 at the endpoints. Chen Ex. 3.3, $\sigma(t)^2 = (M^t - 1)/(M - 1)$ and $\alpha(t)^2 = (M - M^t)/(M - 1)$ for $M > 0$, $M \neq 1$, is Class R and Class Q, not Class S. The $M = 1$ limit is $\sigma(t)^2 = t$, $\alpha(t)^2 = 1 - t$.

The interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023] is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$. The marginal field is affine,

$$b(t, x) = A(t)x + c(t), \quad A = C_t Q^{-1}, \quad c = \dot{m} - Am. \quad (1)$$

The formula $C_t = \dot{\alpha}\Sigma_0 + \dot{\sigma}\Sigma_1$ holds on $(0, 1)$; equivalently $C_t = \frac{1}{2}(\alpha^2)'\Sigma_0 + \frac{1}{2}(\sigma^2)'\Sigma_1$, which extends continuously to $[0, 1]$ for Class R. This is a flow-matching marginal ODE, not a score-based probability-flow ODE in the sense of Song et al. [2021]. If $\Sigma_0 = I$, then Q , Q' , A , and Σ_1 share the eigenbasis of Σ_1 , and mode i obeys $x'_i = a_i(t)x_i + c_i(t)$ with $q_i = \alpha^2 + \lambda_i\sigma^2$ and $a_i = q'_i/(2q_i)$. The covariance drift formula is independent of the mean path $c(t)$. We compare the linear path $(1 - t, t)$ and trigonometric VP $(\cos(\pi t/2), \sin(\pi t/2))$. Following Chen et al. [2025, Definition 3.2], $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2 dt$. For state-independent A this coincides with their path expectation. By definition $\mathcal{R}$ ignores $c(t)$. Error is Gaussian W_2 of the endpoint laws [Gelbrich, 1990]. This is neither pathwise trajectory error nor local truncation error. Equal NFE uses $S = B/s$ steps with $s = 1, 2, 4$ for Euler, Heun, and RK4.

Any regular variance path from 1 to $\lambda > 0$ obeys $\int_0^1 a dt = \frac{1}{2} \ln \lambda$. Hence $\mathcal{R} = \int_0^1 a^2 dt = (\int a)^2 + \text{Var}_t(a) \geq (\frac{1}{2} \ln \lambda)^2$, with equality if and only if a is constant. The one-dimensional minimizer is $q(t) = \lambda^t$, $a \equiv \frac{1}{2} \ln \lambda$, which is Chen et al. Example 3.3. Linear and trigonometric VP are C^1 shared interpolants. In several dimensions, the per-mode schedule $q_i(t) = \lambda_i^t$ yields $\mathcal{R} = (\frac{1}{2} \max_i |\ln \lambda_i|)^2$. When Σ_1 has at least two distinct eigenvalues, that path is not a Class S shared interpolant.

3 A one-dimensional Gaussian ranking counterexample

Assumption 1. Endpoints $N(0, 1)$ and $N(0, 4)$ are independent. Schedules are linear and trigonometric VP. The solver is explicit Heun with NFE 8 (four steps of size 1/4).

Proposition 1. Under the assumption, $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1 > \mathcal{R}_{\text{VP}} = \pi^2/16$, while the Heun factors satisfy $r_{\text{lin}} = 6797469/3559400$ and $0 < r_{\text{VP}} < 187/100$. Hence $W_2^{\text{lin}} < 0.091 < 0.130 < W_2^{\text{VP}}$.

Proof sketch. The linear variance is $q_{\text{lin}}(t) = 1 - 2t + 5t^2$. Completing the square gives the equivalent identities $q_{\text{lin}}(t) = 5[(t - 1/5)^2 + (2/5)^2] = 5(t - 1/5)^2 + 4/5$. The substitution $u = t - 1/5 = (2/5) \tan \theta$ converts $\int a_{\text{lin}}^2$ into $(5/2) \int \sin^2 \theta d\theta$. Limits $\arctan(-1/2)$ and $\arctan 2$ together with $\arctan 2 + \arctan(1/2) = \pi/2$ yield $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1$. The VP integral evaluates to $\pi^2/16$ by a Weierstrass identity. The four linear Heun factors are the rationals $47/52$, $6/5$, $497/370$, and $97/74$, all positive. Exact VP grid values are $a(0) = a(1) = 0$, $a(1/2) = 3\pi/10$, and $a(1/4) = (9 + 15\sqrt{2})\pi/82$, $a(3/4) = (-9 + 15\sqrt{2})\pi/82$. Positivity of $a(3/4)$ is $15^2 \cdot 2 - 9^2 = 369 > 0$. The four-step product lies in $\mathbb{Q}[\pi, \sqrt{2}]$ with nonnegative coefficients, so substituting $\sqrt{2} < 99/70$ and $\pi < 355/113$ is valid and yields $r_{\text{VP}} < 187/100$ by an integer comparison in the anonymous supplement. Interval arithmetic is a cross-check only. $\square$

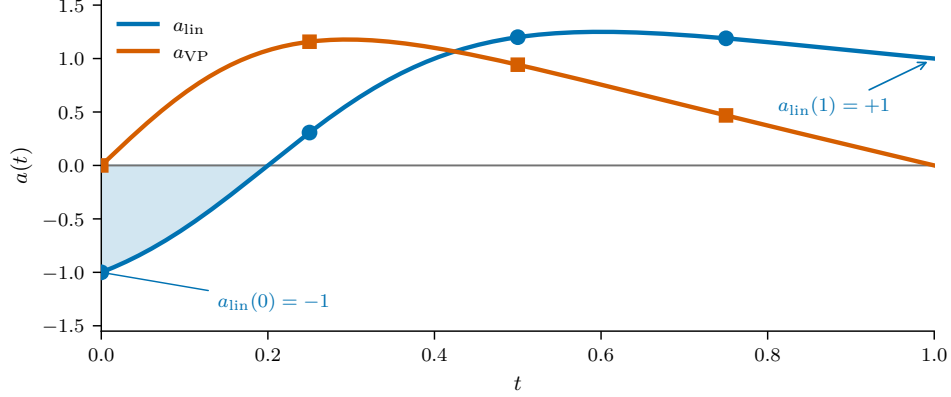


Figure 1: Scalar drifts $a_{\text{lin}}(t)$ and $a_{\text{VP}}(t)$ for independent $N(0, 1)$ and $N(0, 4)$. The shaded region is $a_{\text{lin}} < 0$. Markers are the four Heun nodes at NFE 8. $a_{\text{lin}}(0) = -1 < 0 < a_{\text{lin}}(1) = +1$. Exact evaluation of the closed-form drifts.

77 4 Sign change and transported defects

78 For any independent linear coupling, $a_{\text{lin}}(0) = -1 < 0 < a_{\text{lin}}(1) = +1$. The VP drift vanishes
 79 at the endpoints and does not change sign on $(0, 1)$. Averaging a^2 therefore erases a qualitative
 80 distinction. Fixed-stage Runge–Kutta methods use signed stage evaluations that $\mathcal{R}$ discards; in the
 81 tested schemes, this distinction contributes to the observed inversion.

82 Let $e_n = \sqrt{q(t_{n+1})/q(t_n)}$ and let r_n be the numerical factor. The telescoping identity

$$\prod_n r_n - \prod_n e_n = \sum_n (r_n - e_n) \left(\prod_{j < n} r_j \right) \left(\prod_{j > n} e_j \right) \quad (2)$$

83 is an accounting tool, not the content of the inversion. The content is the leading exact-minus-method
 84 coefficient: $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-a''/12 + a^3/6)h^3$ for the implemented Heun factor. Those
 85 expansions involve different derivatives, signs, and stage locations.

86 5 Four-path census and the inversion region

87 Source $N(0, I)$. Four centered targets: anisotropic Gaussians with condition number 4 in dimensions
 88 2 and 8 (eigenvalues a geometric sequence from $1/2$ to 2); low-rank covariances $FF^\top + 0.05I$ of rank
 89 2 (seed 271828). Because $\mathcal{R}$ uses the spectral norm, the two anisotropic dimensions share the same
 90 $\mathcal{R}$ pair $(0.972296, 0.192270)$, so the grid contains three distinct linear-versus-VP $\mathcal{R}$ comparisons.
 91 Each geometry is run under Euler, Heun, and RK4 at NFE $\{8, 16, 32\}$. Headline $\mathcal{R}$ is the continuous
 92 integral of $\max_i a_i(t)^2$. Ranking inversions were identified in Phase 3; the counts below are a
 93 confirmatory Phase 4 reproduction on configurations frozen before the rerun, not a preregistration
 94 that predates discovery.

95 The per-mode log-covariance path $q_i(t) = \lambda_i^t$ is one of four candidate paths. Among those four paths
 96 on the stated 36-block census it has the smallest $\mathcal{R}$ on every geometry and the smallest Gaussian W_2
 97 in all 36 blocks (Table 1 reports Heun at NFE 8). This finite-census observation does not imply that
 98 a global $\mathcal{R}$ -minimizer minimizes fixed-NFE error. When Σ_1 has at least two distinct eigenvalues,
 99 that path is not a C^1 shared interpolant. Linear versus VP invert in 5 of 12 geometry $\times$ solver cells; 4
 100 of those 12 invert at every tested NFE. As a pairwise comparison within the shared-schedule class,
 101 trigonometric VP versus Chen Ex. 3.3 with $M = \lambda_{\max}$ invert in 9 of 36 blocks and 4 of 12 cells:
 102 anisotropic $d = 2$, Euler at NFE 16 and 32; anisotropic $d = 8$, Euler at NFE 8, 16, and 32; low-rank
 103 $d = 2$, Euler at NFE 8 and Heun at NFE 8, 16, and 32; low-rank $d = 8$, none. The fourteen inverted
 104 linear-versus-VP solver-budget rows are a hierarchical listing of those cells, not a population rate.

105 A one-dimensional sweep $\lambda \in [0.05, 100]$ (Figure 3) locates the linear-versus-VP inversions. At $\lambda =$
 106 4, the inversion is present under Heun at every tested budget NFE $\in \{4, 8, 12, 16, 24, 32, 64, 128, 256\}$,
 107 absent under Euler, and mixed under RK4 (absent at NFE 8 and 12). Euler *does* invert at more

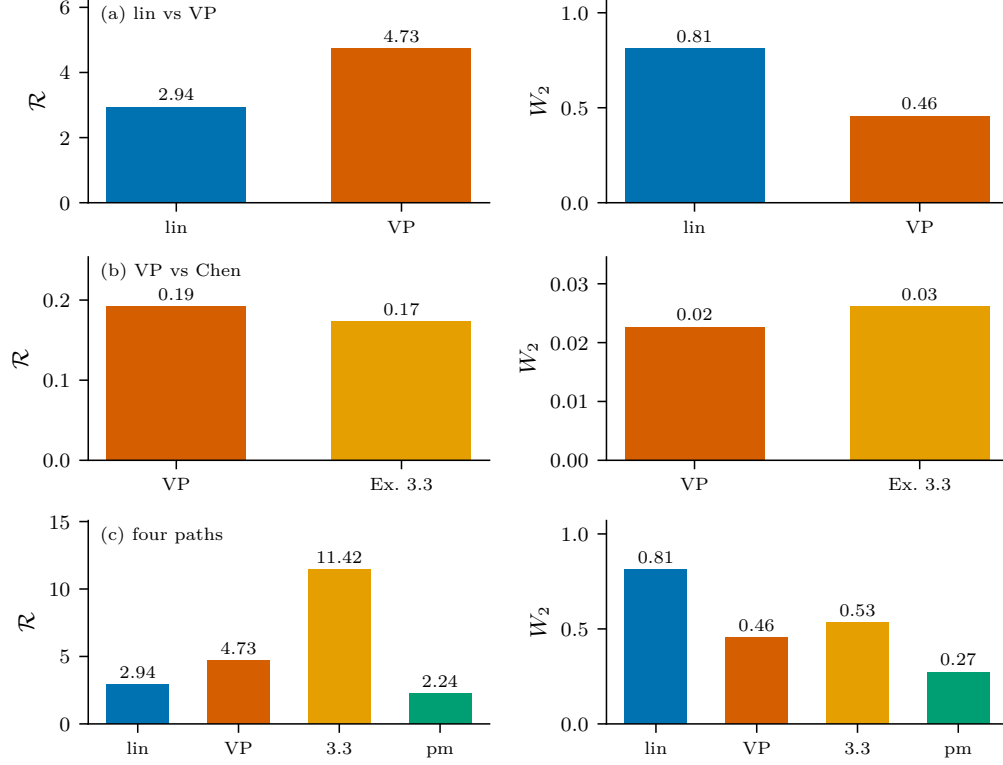


Figure 2: Three comparisons for $\mathcal{R}$. Each row shows continuous $\mathcal{R}$ (left) and Euler NFE 8 Gaussian W_2 (right). (a) linear versus VP on low-rank $d = 8$; (b) VP versus Chen Ex. 3.3 on anisotropic $d = 8$; (c) four paths on low-rank $d = 8$. Counts: 5 of 12 cells, 9 of 36 blocks, 36 of 36 among four candidates. The last does not imply that a global $\mathcal{R}$ -minimizer minimizes fixed-NFE error.

Table 1: Four schedules: linear, trigonometric VP, shared Chen Ex. 3.3 scalar ($\mathcal{R}_{\text{sc}}$, $M = \lambda_{\text{max}}$), and per-mode log-covariance ($\mathcal{R}_{\text{log}}$). $\mathcal{R}$ is continuous; the last column is the W_2 -minimizer at Heun, NFE 8. Among these four candidate paths, per-mode wins Heun-8 on every geometry. This finite-census observation does not imply that a global $\mathcal{R}$ -minimizer minimizes fixed-NFE error. When Σ_1 has at least two distinct eigenvalues, the per-mode path is not a C^1 shared interpolant. The Gaussian drift and W_2 formula are closed form; the regularity integrand is exact, while multimode $\mathcal{R}$ is evaluated deterministically by adaptive quadrature.

family	d	$\mathcal{R}_{\text{lin}}$	$\mathcal{R}_{\text{VP}}$	$\mathcal{R}_{\text{sc}}$	$\mathcal{R}_{\text{log}}$	Heun-8 W_2 min
anisotropic	2	0.972	0.192	0.173	0.120	log-cov
anisotropic	8	0.972	0.192	0.173	0.120	log-cov
low-rank	2	1.246	0.625	0.668	0.380	log-cov
low-rank	8	2.944	4.731	11.419	2.244	log-cov

108 extreme λ on the same sweep. The scalar example is therefore solver-dependent, not an artefact of
109 Heun at budget 8 alone.

Table 2: Strongest centered inversion: low-rank, $d = 8$, Euler, NFE 8. Eigenvalues: 0.05 (six times), 4.64827725629505, 10.675638226546441.

path	$\mathcal{R}$	$\widehat{\mathcal{R}}_{24}$	Gaussian W_2
linear	2.9441044083	2.9476523251	0.8108540111
VP	4.7305438136	4.7295206355	0.4564779075

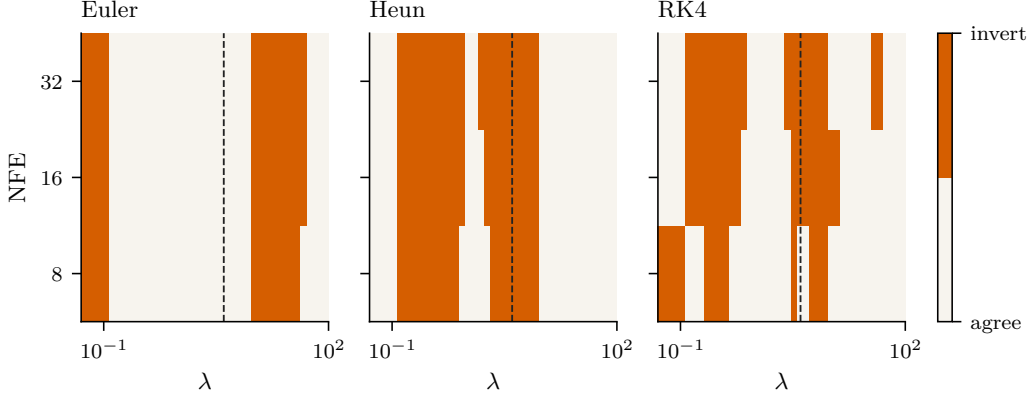


Figure 3: Linear versus VP ranking inversion versus target variance λ . Dark cells: $\mathcal{R}$ and W_2 disagree. The dashed line is $\lambda = 4$ (Proposition 1).

6 A grid-aware Class S Euler construction

The 36 of 36 count is a finite census among four candidates. It does not imply that a global $\mathcal{R}$ -minimizer minimizes fixed-NFE error. The following construction makes that fence explicit. Fix $L > 0$ and $N \geq 1$, set $h = 1/N$ and $\varepsilon_N = N(e^{L/N} - 1) - L > 0$, and define $a_0(t) = L$ and $a_{1,N}(t) = L + \varepsilon_N \cos(2\pi Nt)$. Both exact endpoints of $x' = a(t)x$, $x(0) = 1$ equal e^L . The constant field a_0 uniquely minimizes $\int_0^1 a^2$ among scalar fields with integral L . The oscillatory field has strictly larger regularity $\int a_{1,N}^2 = L^2 + \varepsilon_N^2/2$. Left-endpoint Euler is exact for $a_{1,N}$ and has strictly positive endpoint error for a_0 . The embedding $q_k = \exp(2 \int_0^t a_k)$, $\alpha_k = \sqrt{q_k} \cos(\pi t/2)$, $\sigma_k = \sqrt{q_k/e^{2L}} \sin(\pi t/2)$ lies in Class S. The field $a_{1,N}$ is chosen after the grid; this is not the mechanism of Proposition 1.

7 Related work

Flow matching and interpolants supply the Gaussian family [Lipman et al., 2023, Albergo et al., 2025, Liu et al., 2023, Tong et al., 2024]. Lipschitz-guided design proposes the selection criterion tested [Chen et al., 2025]. Solver-specific schedules and exact linear parts are documented in diffusion practice [Sabour et al., 2024, Karras et al., 2022, Lu et al., 2022, Zhang and Chen, 2023]. Stéphanovitch [2026] prove sharp Lipschitz regularity of canonical flow-matching fields and Euler-type W_2 rates; that is a bound, not a pairwise ranking of two interpolants at equal NFE.

Tao and Choi [2026] decompose $\nabla v = S + \Omega$ and control Euler error by the logarithmic norm $\mu_2 = \lambda_{\max}(S)$, which is signed, whereas $\mathcal{R}$ integrates $\|A\|_2^2$. In one dimension $\mu_2(a) = a$ and $\|A\|_2 = |a|$. Their Theorem 5 states that McCann displacement interpolation has vanishing material derivative, hence Euler is order 2 in the Eulerian analysis; Remark 7 records the stronger Lagrangian exactness observed on that interpolant. Neither statement applies to the independent-coupling linear or VP path studied here. The present paper supplies a rationally certified Gaussian instance of ranking failure for the unsigned average $\|A\|_2^2$. A counterexample to μ_2 itself is outside our scope. Benton et al. [2024] bound W_2 of a learned flow; that is an error bound, not a pairwise ranking. Consistency models [Song et al., 2023] distill few-step maps.

8 Limitations

The systems are commuting affine Gaussian interpolants, three fixed-step Runge–Kutta methods, and small equal-NFE budgets. No learned neural field is evaluated. Chen et al. prove no discretization-error bound and do not claim a universal pairwise ranking at fixed NFE. As a pairwise comparison within the shared-schedule class, VP versus Chen Ex. 3.3 with $M = \lambda_{\max}$ still invert in 9 of 36 blocks. Among four candidate paths, the per-mode path wins 36 of 36 blocks on this census; that does not imply that a global $\mathcal{R}$ -minimizer minimizes fixed-NFE error. The Gaussian family is a probe in which the drift and W_2 formula are closed form; in a learned field the linear part is not exact, so exponential integrators that solve an affine scalar mode exactly are not a drop-in substitute. Cell counts are hierarchical descriptive summaries, not population rates.

9 Conclusion

Averaged squared Jacobian regularity can misrank equal-NFE Gaussian W_2 already for two C^1 interpolants of independent $N(0, 1)$ and $N(0, 4)$. The Heun calculation is exact on the linear path and rationally enclosed on trigonometric VP. The same split appears as a pairwise comparator (5 of 12 geometry×solver cells), as a pairwise shared-schedule comparison (9 of 36 blocks), and as a four-candidate census (36 of 36) that does not promote A_2 to a global few-step optimality theorem.

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